

CSF 3125 OPERATING SYSTEM [3-0-0-3]

What Operating Systems Do, Operating System Structure, Operating System Operations, Process Management, Memory Management, and Storage Management. Operating System Services, User and Operating System Interface, System Calls, Types of System Calls, System Programs, Operating System Structure, Virtual Machines, System Boot. Overview, Process Scheduling, Operations on Processes, Interprocess Communication. Overview, Multithreaded Models, Thread Libraries. Background, The Critical Section Problem, Peterson's Solution, Synchronization Hardware, Mutex Locks, Semaphores. Basic Concepts, Scheduling Criteria, Scheduling, Thread Scheduling, Linux Scheduling. System Model, Deadlock, Characterization, Methods for handling deadlocks, Deadlock prevention, Deadlock Avoidance. Logical Versus Physical Address Space, Swapping, Contiguous Memory Allocation, Segmentation, Paging, Structure of Page Table. Background, Demand Paging, Copy-On-Write, Page Replacement, Allocation of Frames, Thrashing. Disk Structure, Disk Scheduling, Swap-Space Management. File Concept, Access Methods, Directory and Disk Structure, File System Mounting. Goals of Protection, Principles of Protection, Domain of Protection, Access Matrix Implementation of Access Matrix.

References Books:

1. A. Silberschatz, P. B. Galvin and G. Gagne, Operating System Concepts, (9e), Wiley and Sons (Asia) Pte Ltd, 2016
2. D.M. Dhamdhere, Operating Systems A Concept based Approach,(2e), Tata McGraw-Hill 2006
3. Deitel & Deitel , Operating systems, (3e), Pearson Education, India. 2018.
4. Neil Matthew, Richard Stones, Beginning Linux Programing(4e), Wrox, 2007
5. Andrew S Tanenbaum, Herbert Bos, Modern Operating systems. Pearson Education India, 2018